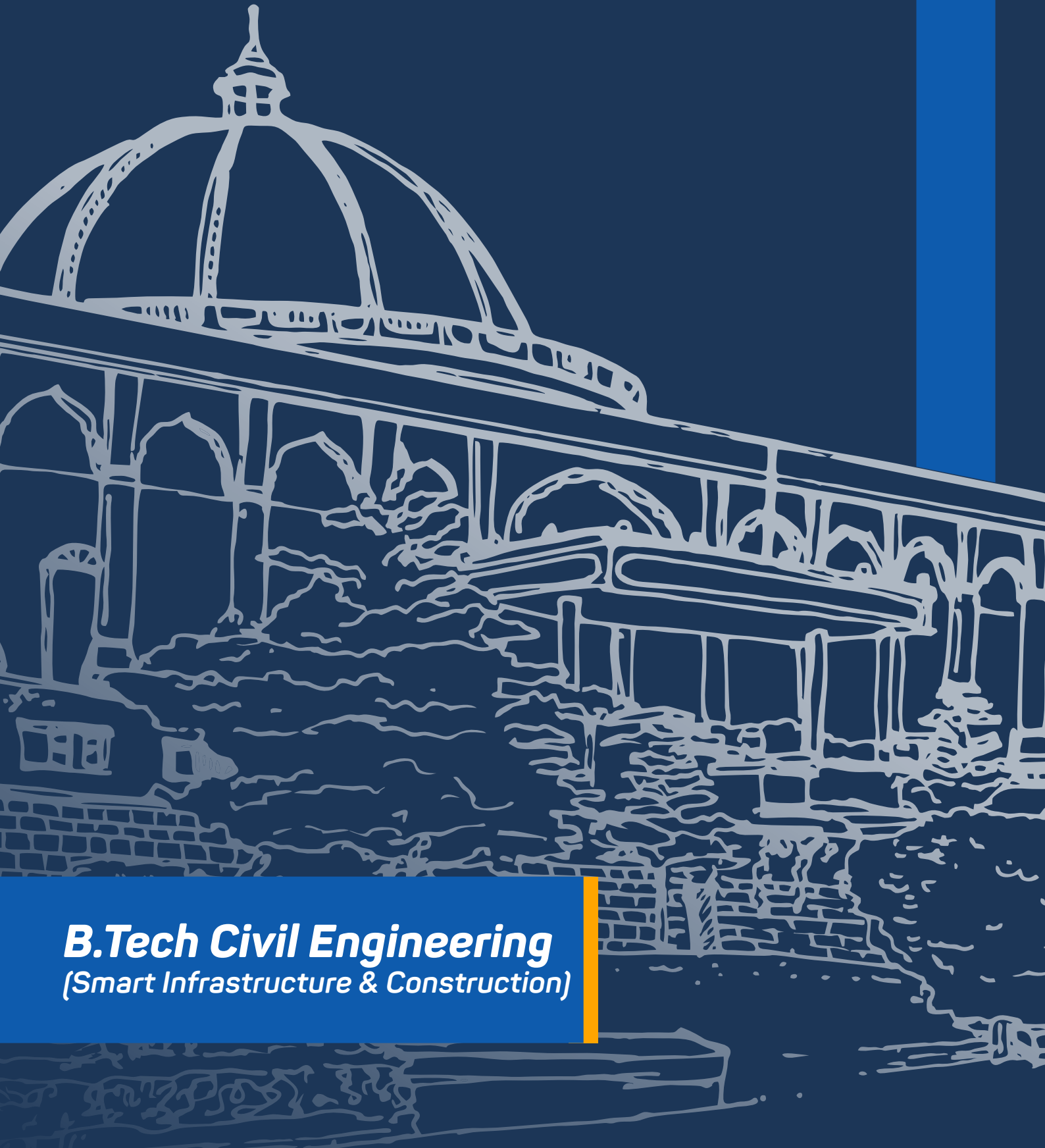




B.Tech Civil Engineering
(Smart Infrastructure & Construction)

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	Department of Civil Engineering
Programme Name	B.Tech Civil Engineering (Smart Infrastructure & Construction)

Semester	Odd (I)	Even (II)	Total Credits
First Year	22	22	44
Second Year	22	22	44
Third Year	21	22	43
Fourth Year	19	15	34

+ + + + + + + + + + + + + + + **COURSE BASKET** + + + + + + + + + + + + + + +

| Course Type | Description |
|----------------------|--|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU |
| Programme Electives | Open electives under the programme allow students to specialise in a particular area connected to their major. |
| University Core | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core. |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| I | UC | Indian Constitution | 1 |
| I | UC | Environment and Sustainability | 1 |
| I | UC | Yoga - I | 1 |
| I | UC | Social Leadership Development Program | 1 |
| I | UC | Financial Literacy | 1 |
| I | PF | Linear Algebra & Differential Calculus | 3 |
| I | PF | Engineering Chemistry | 3 |
| I | PF | Engineering Physics | 3 |
| I | PF | Ideas and Innovation in Manufacturing | 1 |
| I | PM | Engineering Mechanics | 4 |
| I | PF | Programming and Problem solving | 3 |
| | | Total Credits: | 22 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| II | UC | Yoga - II | 1 |
| II | UC | Co-creation | 1 |
| II | UC | AI for everyone | 2 |
| II | UC | Foundation of Peace | 2 |
| II | UC | Indian Knowledge System (General) | 2 |
| II | UC | Sports | 1 |
| II | PF | Integral Calculus | 3 |
| II | PF | Biology for Engineers | 2 |
| II | PF | Engineering Graphics | 3 |
| II | PF | Basics of Electrical and Electronics Engineering | 3 |
| II | PM | Surveying Field Practices | 2 |
| | | Total Credits: | 22 |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| III | UC | Spiritual and Cultural heritage: Indian Experience | 2 |
| III | UC | Research Innovation Design Entrepreneurship (RIDE) | 1 |
| III | UE | University Electives - I | 3 |
| III | PF | Calculus and Numerical Methods | 4 |
| III | PM | Strength of Materials | 3 |
| III | PM | Fluid Mechanics | 3 |
| III | PM | Smart Infrastructure Engineering | 2 |
| III | PF | IOT for Smart and Intelligent buildings | 1 |
| III | Program Capstone Project/
Seminar and Internships | Building Planning and CAD | 3 |
| | | Total Credits: | 22 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| IV | UC | Rural Immersion | 1 |
| IV | UC | Life Transformation Skills | 1 |
| IV | UE | University Electives - II | 3 |
| IV | PF | Indian Knowledge System (Sci. & Tech). | 2 |
| IV | PF | Probability and Statistics for Civil Engineers | 4 |
| IV | PM | Surveying and Geomatic Engineering | 3 |
| IV | PM | Structural Analysis I | 3 |
| IV | Program Capstone Project/
Seminar and Internships | Design Thinking | 2 |
| IV | PM | Concrete Technology | 3 |
| | | Total Credits: | 22 |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| V | UC | Managing Conflicts Peacefully: Tools and Techniques | 2 |
| V | UE | University Electives - III | 3 |
| V | PE | Program Elective - I | 4 |
| V | PM | Design of Steel Structures | 3 |
| V | PM | Project Management | 3 |
| V | PM | Structural Analysis II | 3 |
| V | Program Capstone Project/Seminar and Internships | Civil Engineering Softwares | 2 |
| V | Program Capstone Project/Seminar and Internships | Data Science for Engineers | 1 |
| | | Total Credits: | 21 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| VI | UC | National Academic Immersion Program | 2 |
| VI | PE | Program Elective - II | 4 |
| VI | PF | Soil Mechanics | 3 |
| VI | PF | Transportation Engineering | 3 |
| VI | PM | Design of Reinforced Concrete Structures | 3 |
| VI | PM | Water and Waste Water Engineering | 3 |
| VI | Program Capstone Project/Seminar and Internships | AIML in Infrastructure Engineering | 1 |
| VI | Program Capstone Project/Seminar and Internships | Mini Project | 1 |
| VI | Program Capstone Project/Seminar and Internships | BIM in Civil Engineering | 2 |
| | | Total Credits: | 22 |
| | | | |

Semester - VII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| VII | PE | Program Elective - III | 4 |
| VII | PM | Foundation Engineering | 3 |
| VII | PM | Pavement Material Charecterization | 1 |
| VII | PM | Dams and Hydraulics Structures | 3 |
| VII | PM | Quantity Surveying and Estimation | 3 |
| VII | Program Capstone Project/Seminar and Internships | Computer aided prestressed concrete design | 3 |
| VII | Program Capstone Project/Seminar and Internships | Capstone project | 2 |
| | | Total Credits: | 19 |

Semester - VIII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--------------------------------------|---------------|
| VIII | PE | Program Elective - IV | 4 |
| VIII | Program Capstone Project/Seminar and Internships | Hydrology and Irrigation Engineering | 3 |
| VIII | Program Capstone Project/Seminar and Internships | AI in Civil Engineering | 2 |
| VIII | Program Capstone Project/Seminar and Internships | Internship | 6 |
| | | Total Credits: | 15 |

Program Electives

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| V | PE - I | Structural Health Monitoring | 4 |
| V | PE - I | Structural Audit and Retrofitting | 4 |
| V | PE - I | Finite Element Methods | 4 |
| V | PE - I | Advanced construction methods and automation | 4 |
| V | PE - I | Heavy Machinery in Construction Project | 4 |
| V | PE - I | International Contracting | 4 |
| V | PE - I | Solid Waste Management | 4 |
| V | PE - I | Air Pollution and Control | 4 |
| V | PE - I | GIS and Remote Sensing | 4 |
| V | PE - I | Infrastructure Development and Construction Technology | 4 |
| VI | PE - II | Matrix Methods | 4 |
| VI | PE - II | Earthquake Engineering | 4 |
| VI | PE - II | Fastening systems in concrete construction (HILTI) | 4 |
| VI | PE - II | Construction safety and engineering | 4 |
| VI | PE - II | Total Quality Management and Management Information Systems | 4 |
| VI | PE - II | Tenders and contracts | 4 |
| VI | PE - II | Traffic Flow Models and Simulation | 4 |
| VI | PE - II | Architecture and Town Planning | 4 |
| VII | PE - III | Advanced Design of Structures | 4 |
| VII | PE - III | Ferrocement Design and Technology | 4 |
| VII | PE - III | Analysis and design of transportation Infrastructure | 4 |
| VII | PE - III | Smart City Planning and Development | 4 |
| VII | PE - III | Environmental Impact Assessment of Infrastructure projects | 4 |
| VII | PE - III | Advanced Wastewater systems | 4 |
| VII | PE - III | Planning and Design of Non-Motorized traffic | 4 |
| VIII | PE - IV | Advanced Geotechnical Engineering | 4 |
| VIII | PE - IV | Advanced Foundation Engineering | 4 |
| VIII | PE - IV | Real Estate Valuation | 4 |
| VIII | PE - IV | Construction Management | 4 |
| VIII | PE - IV | Sustainable field Practices | 4 |
| VIII | PE - IV | Tunnel Engineering | 4 |
| VIII | PE - IV | Subsea Engineering | 4 |
| VIII | PE - IV | Sustainable transportation systems | 4 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.